

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103000740 - Correctness By Construction

DEGREE PROGRAMME

10AK - Master Universitario En Software Y Sistemas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	103000740 - Correctness By Construction
No of credits	6 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	10AK - Master Universitario en Software y Sistemas
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Manuel Carro Liñares (Subject coordinator)	2303	manuel.carro@upm.es	M - 13:00 - 15:00 Tu - 13:00 - 15:00 W - 13:00 - 15:00 Please note that the office hours may change during the course. Please get in touch with the instructor to get an appointment.

Manuel De Hermenegildo Salinas	2212	manuel.hermenegildo@upm. es	Sin horario. Please get in touch with the instructor to get an appointment.
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* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Concurrent systems and concurrent programming
- Declarative programming
- First-order logic
- Programming experience (minimum 2 years)
- Formal proofs
- Reasoning about properties of algorithms

4. Skills and learning outcomes *

4.1. Skills to be learned

CEM1 - Identificar, a partir del estado de la cuestión, la presencia de problemas de investigación relacionados con la concepción, la construcción, el uso y la evaluación de sistemas sociotécnicos complejos que hagan un uso intensivo de software

CEM4 - Analizar y evaluar los diferentes paradigmas y enfoques de ingeniería de construcción y gestión de sistemas basados en software.

CEM5 - Aportar soluciones a aquellos problemas abiertos relacionados con el ámbito de aplicación y los métodos, técnicas y herramientas de Verificación y Validación de Software

CG13 - Apreciación de los límites del conocimiento actual y de la aplicación práctica de la tecnología más reciente.

CG7 - Especificación y realización de tareas informáticas complejas, poco definidas o no familiares

4.2. Learning outcomes

RA96 - Acquaintance with the formalisation of programming language syntax

RA122 - RA-AV-8: Be able to use existing tools for formal program verification.

RA123 - RA-AV-11: Be able to give formal specifications of the expected results of programs.

RA91 - Acquaintance with design requirements and implementation requirements.

RA94 - Effective use of rigorous software development techniques.

RA97 - Acquaintance with the formalisation of programming language semantics

RA98 - Ability to reason about recursion and perform proofs by induction

RA99 - Comprender los fundamentos del paradigma de computación orientada a servicios y entender el lugar que ocupa y las ventajas que aporta en relación con otros paradigmas existentes

RA93 - Knowledge of languages which ease the application of the aforementioned techniques.

RA124 - RA-AV-12: Understand, at the level of a user, the automatic demonstration techniques more widely used in the tools for program verification.

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Software is becoming increasingly complex and responsible for critical tasks. Any technology aimed at ensuring the reliability and quality of software will be increasingly relevant, if not utterly necessary.

Only rigorous (e.g., mathematically sound) approaches can certify software with the highest possible assurance. These approaches include, among others, the use of specification languages, high-level programming languages (including equational, functional, and logic languages), the use of model checking and deductive verification, language-based approaches often interacting with theorem provers.

In this course we will give a hands-on introduction to rigorous software development methods that follow a *correctness-by-construction* approach. While the course is not heavy in theory, everyone is expected to have a good understanding of first-order logic and programming experience.

5.2. Syllabus

1. Introduction to Formal Methods: Proving Programs Correct
2. Fundamentals of Formal Methods: Specification, First-Order Logic, Proofs, Programs
3. Event-B Basics and the Rodin Tool
4. Sequential Systems
5. Event B: Mathematical Toolkit and Applications
6. Reactive Systems: Concurrency and Distribution

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Introduction to formal methods and correctness by construction Duration: 01:30 Lecture Sample cases of formal development Duration: 01:30 Cooperative activities			
2	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
3	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			Homework Individual work Progressive assessment Not Presential Duration: 04:00
4	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
5	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
6	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			Homework Individual work Progressive assessment Not Presential Duration: 04:00

7	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
8	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
9	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			Homework Individual work Progressive assessment Not Presential Duration: 08:00
10	Quizzes Duration: 01:00 Problem-solving class Event-B and related topics Duration: 02:00 Lecture			
11	Presentation of term project Duration: 01:00 Additional activities Event-B and related topics Duration: 02:00 Lecture			Term project Group work Progressive assessment Not Presential Duration: 20:00
12	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
13	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
14	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			

15	Presentaciones de trabajo en grupo Duration: 03:00 Additional activities			Presentation and defense of group projects Group presentation Progressive assessment Presential Duration: 03:00
16				
17				Final regular exam Written test Global examination Presential Duration: 03:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
3	Homework	Individual work	No Presential	04:00	20%	2 / 10	CEM1 CG7 CG13 CEM4 CEM5
6	Homework	Individual work	No Presential	04:00	20%	2 / 10	CEM1 CG7 CG13 CEM4 CEM5
9	Homework	Individual work	No Presential	08:00	20%	2 / 10	CEM1 CG7 CG13 CEM4 CEM5
11	Term project	Group work	No Presential	20:00	%	4 / 10	CEM1 CG7 CG13 CEM4 CEM5
15	Presentation and defense of group projects	Group presentation	Face-to-face	03:00	40%	4 / 10	CEM1 CG7 CG13 CEM4 CEM5

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Final regular exam	Written test	Face-to-face	03:00	100%	5 / 10	CEM1 CG7 CG13 CEM4 CEM5

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Extra final exam	Written test	Face-to-face	03:00	100%	5 / 10	CEM1 CG7 CG13 CEM4 CEM5

7.2. Assessment criteria

- No mandatory activities are necessary to pass via the final exams
- The minimum grade to pass the course is 5 over 10 (either when it is calculated as the weighted sum of individual homework or when it is the grade of a single comprehensive exam).
- The topics covered in the different homework assignments cannot be tested separately in the final exam, as they are deeply intertwined and are not isolated units of knowledge.
- The global exams, both the regular and the extraordinary ones, will be in person.
- Copying from any source (either textbooks, the Internet, another student, or any other source) with or without the permission of the author of the source, as well as other types of academic fraud, can lead to a 'fail' grade in the course and / or being reported to the academic authorities, who will decide whether to take additional authoritative measures. In particular, in case of non-ethical or fraudulent behavior, the Law 3/2022 of February 24th will be applied, as well as the corresponding UPM regulations. Article 12 and 14 of Law 3/2022 states that a serious fault may mean, among other outcomes, failing the corresponding sitting.
- There are no learning blocks whose earned grades can be carried over to future academic courses.
- Failure to deliver a homework assignment at the time and in the form stated by the instructor(s) may result in a failure for that exercise.
- For progressive evaluation: if for any reason it is not possible to prepare / hand out some homework assignment, its weight in the final grade will be split among the rest of the homework exercises in such a way that the relative weight of the rest of the assignments, when compared with each other, will be the same they had before removing the homework that could not be handed out.
- The term project handed out in week 11 is presented in week 15 to the classroom by every team, instead of just being sent to the professor and assessed independently. Therefore the evaluation activity in week 15 is the closing milestone of the term project, and is part of it. That is the reason why it does not have a separate associated weight.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Lawrence Paulson's class notes	Bibliography	Lawrence Paulson's Logic and Proof are the course notes of the author for a Logic course in Cambridge. Highly recommended, as they are both rigorous and very concise. They provide very good background material for both parts of the course.
Logic in Computer Science (Huth and Ryan)	Bibliography	A very good book on the use of logic in computer science is Logic in Computer Science, by Huth and Ryan. The Computer Science School should have several copies. There may be electronic copies on the Internet, if possible of the second edition.
http://wiki.event-b.org/	Web resource	Central Event-B site
Modeling in Event-B: System and Software Engineering, by Jean-Raymond Abrial.	Bibliography	The reference book for Event B, with plenty of worked examples.

9. Other information

9.1. Other information about the subject

This course will be given in English. Please note that in case Spanish appears as the course language in the general description, that would be a clerical mistake.